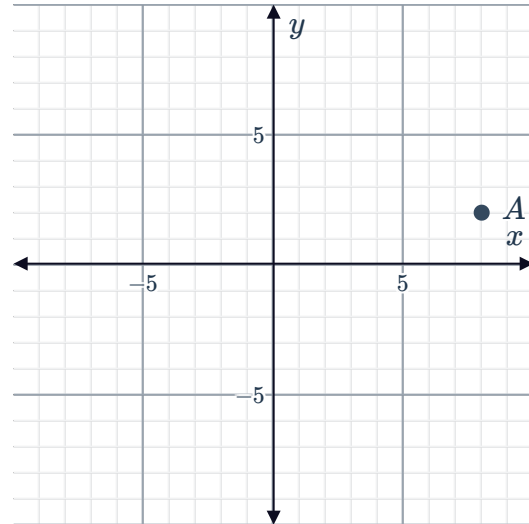


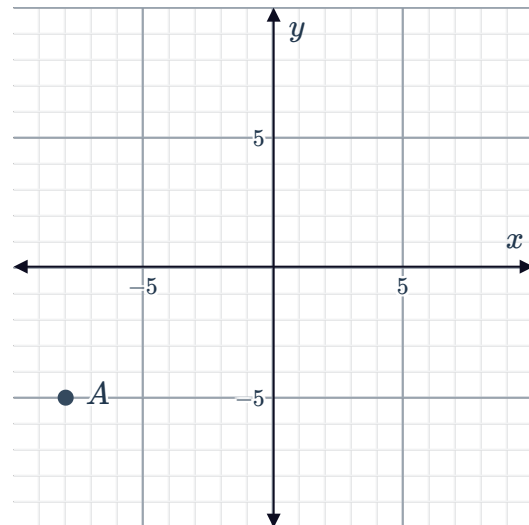
Compositions of transformations



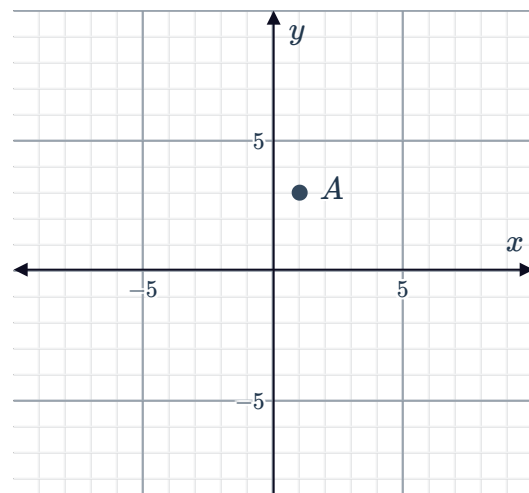
- 1 Point A on the coordinate plane is to undergo two successive transformations.
- a Plot A' , which is the result when A is translated 3 units left.
 - b Plot A'' , which is the result when A' is rotated 90° clockwise about the origin.



- 2 Point A on the coordinate plane is to undergo two successive transformations.
- a Plot A' , which is the result when A is translated 2 units upwards.
 - b Plot A'' , which is the result when A' is rotated 90° clockwise about the origin.

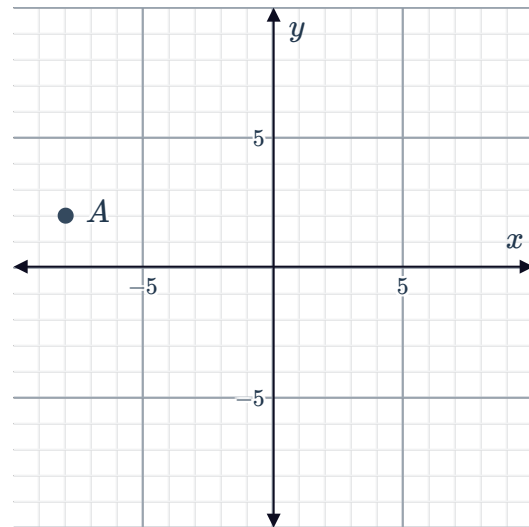


- 3 Point A on the coordinate plane is to undergo two successive transformations.
- a Plot A' , which is the result when A is translated 3 units right and 5 units down.
 - b Plot A'' , which is the result when A' is rotated 90° clockwise about the origin.

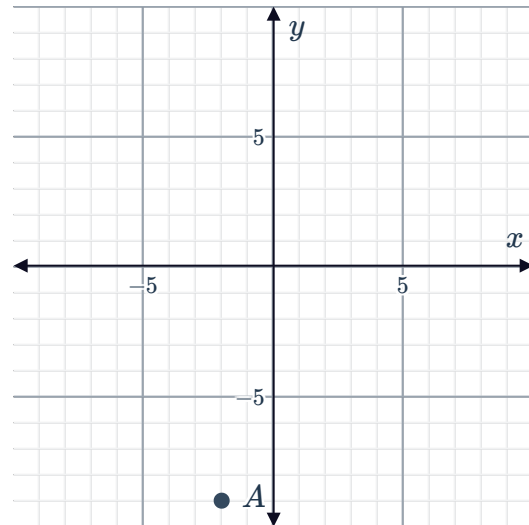




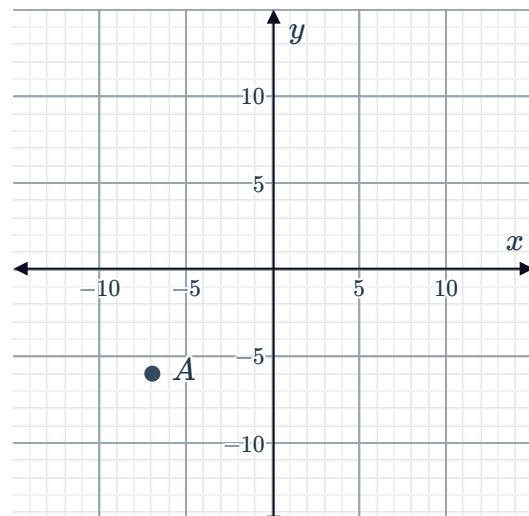
- 4 For the point on the coordinate plane, plot A' that will result from a translation of 5 units right followed by a rotation of 90° counterclockwise about the origin.



- 5 For the point on the coordinate plane, plot A' that will result from a translation of 3 units upwards followed by a rotation of 90° counterclockwise about the origin.



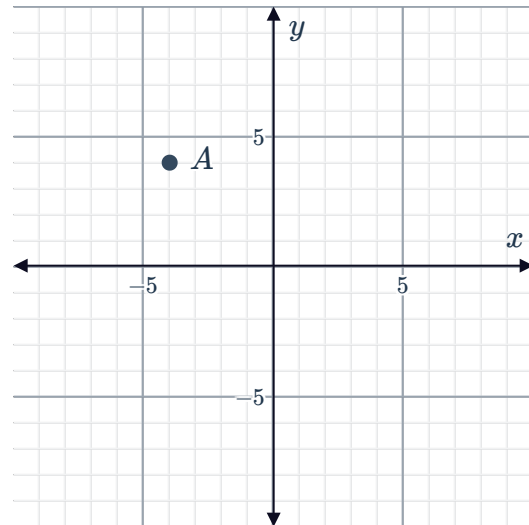
- 6 For the point on the coordinate plane, plot A' that will result from a translation of 5 units left and 5 units down, followed by a rotation of 90° counterclockwise about the origin.





7 A is marked on the following coordinate plane.

- a If A is translated 8 units down to make B , determine the coordinates of B .
- b Across what line could A be reflected so that it overlaps B ?



- 8
- a For any point (a, b) on the coordinate plane, determine the coordinates of the point after a rotation of 180° counterclockwise about the origin.
 - b True or False: A rotation of 180° counterclockwise about the origin is the same as reflecting the point across both the x - and y -axes.

9 $A(-9, 7)$ is rotated 90° clockwise about the origin to form B .

- a Determine the coordinates of B .
- b B is then translated to the left so that it lies on the same vertical line as point A . How many units to the left is point B translated?

10 Determine the coordinates of the transformed point for each of the following:

- a The point $(-9, -1)$ undergoes translation of 2 units down followed by a rotation of 90° clockwise about the origin.
- b The point $(9, 2)$ undergoes a translation of 2 units up followed by a rotation of 90° counterclockwise about the origin.
- c The point on the coordinate plane $(9, -2)$ undergoes a translation of 5 units left and 5 units up, followed by a rotation of 90° clockwise about the origin.
- d The point $(-2, 6)$ undergoes a translation of 5 units right and 4 units up, followed by a rotation of 90° counterclockwise about the origin.

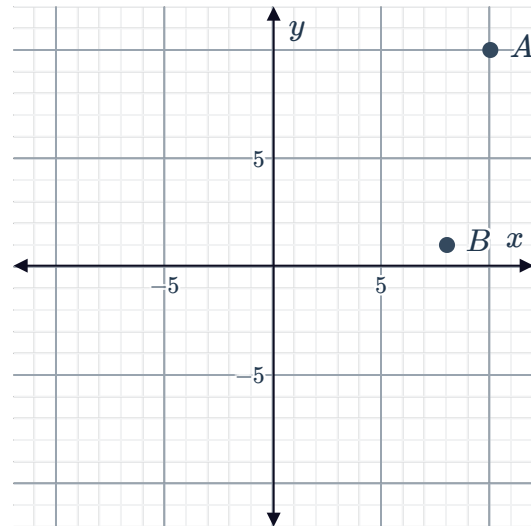
11 $A(6, -1)$ is first rotated 180° about the origin, then it is reflected across the x -axis. This produces the point A' .



- a Determine the coordinates of A' .
- b Determine whether the following transformations also takes A to A' .
 - i A reflection across the y -axis.
 - ii A 90° clockwise rotation about the origin.
 - iii A vertical translation.
 - iv There is no other way to describe the transformation.

12 Consider the two points, A and B on the coordinate plane.

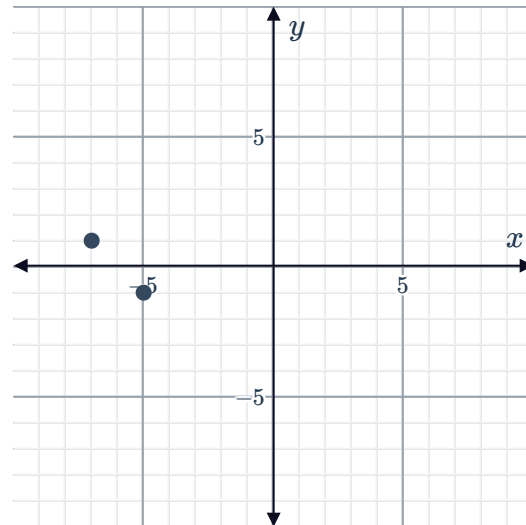
- a Plot the points that would result when A and B are rotated 180° about the origin.
- b Plot the points that would result when A and B are first rotated 180° about the origin, and then reflected across the x -axis.
- c Determine whether the following could be use to describe the transformation in part (b).
 - i There is no other way to describe the transformation.
 - ii A reflection across the y -axis.
 - iii A horizontal translation.
 - iv A vertical translation.



13 Consider the two points on the coordinate plane



- a Plot the points that would result when the original points are rotated 180° about the origin and reflected across the y -axis.
- b Determine whether the following could describe the transformation.
 - i A horizontal translation.
 - ii There is no other way to describe the transformation.
 - iii A vertical translation.
 - iv A reflection across the x -axis.

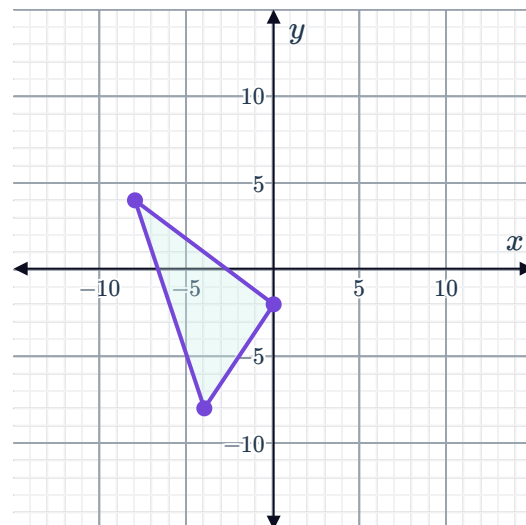


14 Consider the points $A(-10, -8)$ and $B(-7, -7)$.

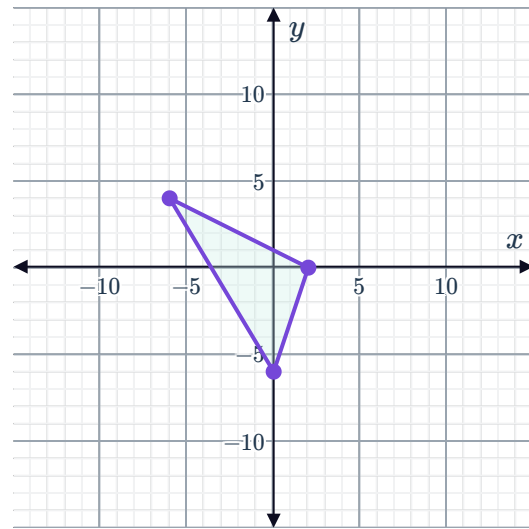
- a Determine the coordinates of A' and B' that would result when the original points are rotated 180° about the origin and reflected across the y -axis.
- b Determine whether the following could also describe the transformation.
 - i There is no other way to describe the transformation.
 - ii A reflection across the x -axis.
 - iii A vertical translation.
 - iv A horizontal translation.

15 The given triangle is to undergo two transformations.

- a Plot the triangle that results from reflecting the given triangle across the x -axis.
- b Plot the triangle that results when the triangle from part a is translated 4 units right.

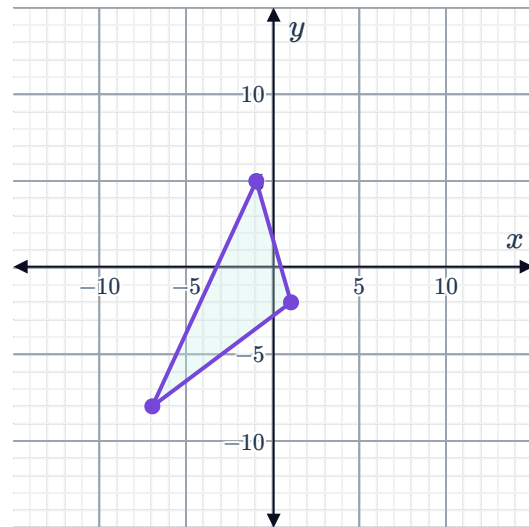


- 16 Plot the triangle that will result when the given triangle is reflected across the x -axis and translated 3 units to the right.

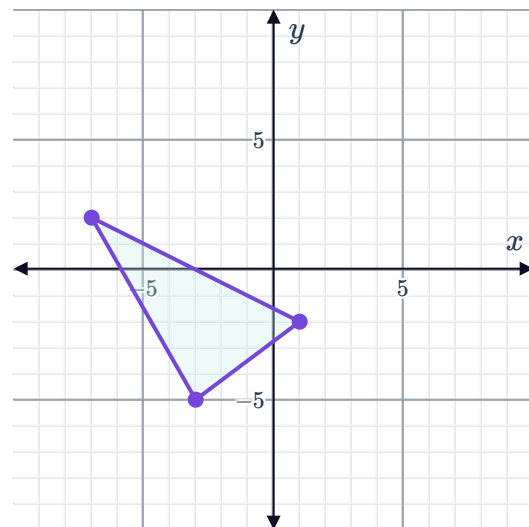


- 17 The given triangle is to undergo two transformations.

- a Plot the triangle that results from reflecting the given triangle across the y -axis.
- b Plot the triangle that results when the triangle from part a is translated 3 units right.



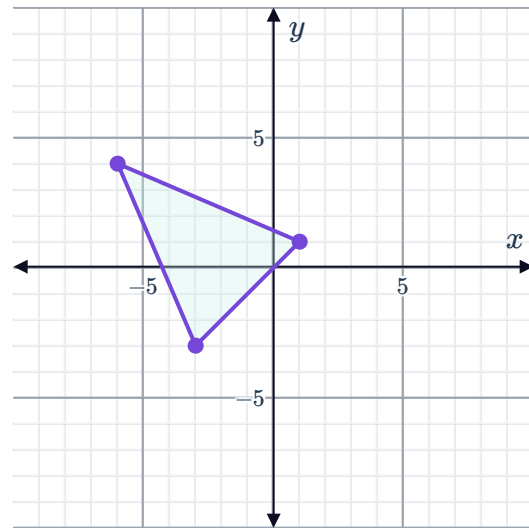
- 18 Plot the triangle that will result when the original triangle is reflected across the y -axis and translated 3 units down.



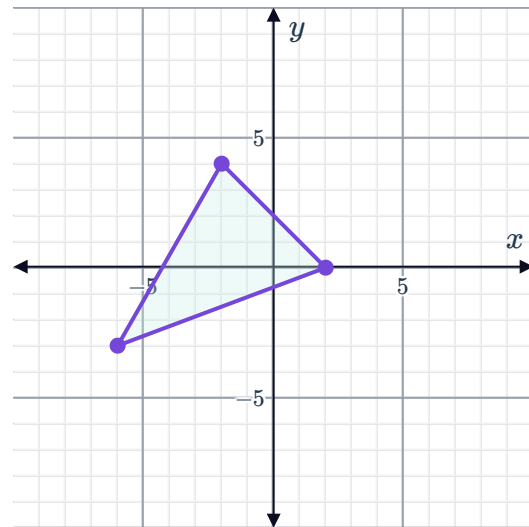
19 The given triangle is to undergo two transformations.



- a Plot the triangle that results from reflecting the given triangle across the x -axis.
- b Plot the triangle that results when the triangle from part b is translated 2 units right and 4 units up.

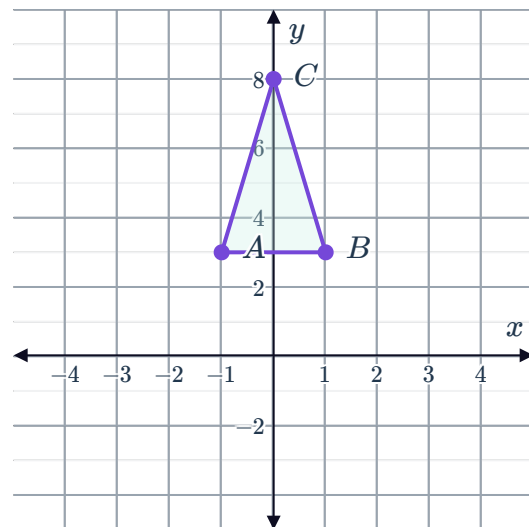


20 Plot the triangle that will result when the original triangle is reflected across the y -axis, and translated 2 units left and 3 units down.



21 $\triangle ABC$ is to undergo two separate reflections.

- a Plot $\triangle A'B'C'$, the result of reflecting $\triangle ABC$ across the line $y = x$.
- b Now suppose $\triangle A'B'C'$ is reflected across the y -axis to form $\triangle A''B''C''$. What single transformation would overlap $\triangle ABC$ onto triangle $\triangle A''B''C''$?



- 22** Points $A(-7, 5)$, $B(-5, -8)$, and $C(10, -10)$ are the vertices of a triangle.



Determine the coordinates of A' , B' , and C' that result from reflecting the triangle across the x -axis and translating it 4 units right.

- 23** Points $A(-10, 4)$, $B(5, -5)$, and $C(5, -2)$ are the vertices of a triangle.

Determine the coordinates A' , B' , and C' that result from reflecting the triangle across the x -axis and translating it 3 units down.

- 24** Points $A(4, -9)$, $B(4, 3)$, and $C(-8, 2)$ are the vertices of a triangle.

Determine the coordinates A' , B' , and C' that result from reflecting the triangle across the y -axis and translating it 5 units right.

- 25** Points $A(8, -4)$, $B(5, 6)$, and $C(9, 9)$ are the vertices of a triangle.

Determine the coordinates A' , B' , and C' that result from reflecting the triangle across the y -axis and translating it 3 units up.

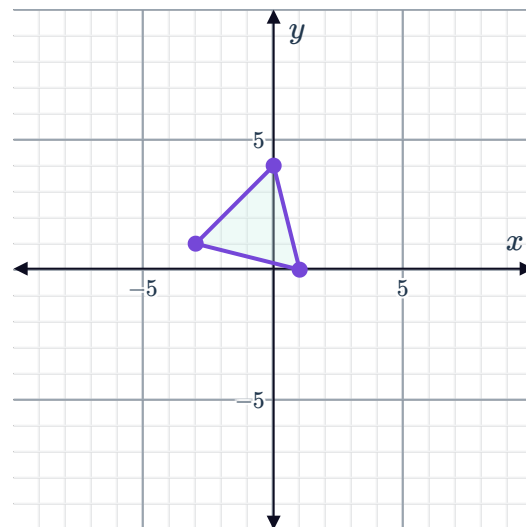
- 26** Points $A(-1, -9)$, $B(-6, -8)$, and $C(8, 9)$ are the vertices of a triangle.

Determine the coordinates A' , B' , and C' that result from reflecting the triangle across the x -axis, and translating it 5 units right and 3 units down.

- 27** Points $A(-5, -7)$, $B(4, 4)$, and $C(9, 1)$ are the vertices of a triangle.

Determine the coordinates A' , B' , and C' that result from reflecting the triangle across the y -axis, and translating it 3 units right and 5 units up.

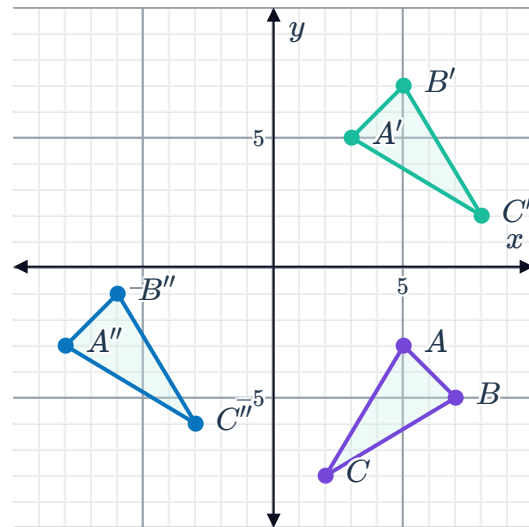
- 28** Consider the triangle shown on the coordinate grid.
Plot the triangle that results when the first triangle undergoes the transformation $(x, y) \rightarrow (x - 3, y + 4)$.



29 Determine whether the following sequences of transformations map $\triangle ABC$ to $\triangle A''B''C''$.

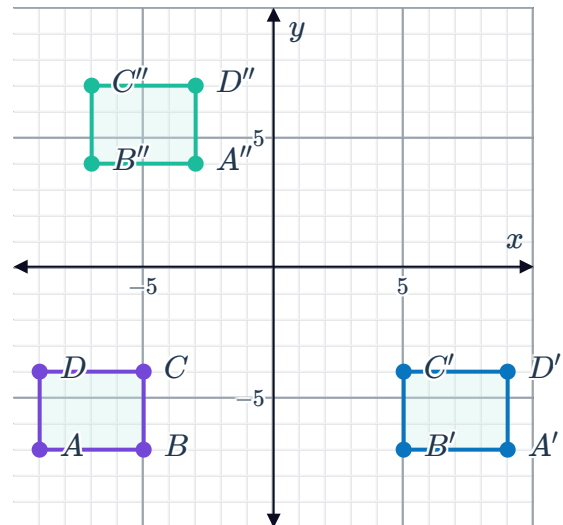


- a A reflection across the x -axis, and a translation 11 units left and 8 units down.
- b A translation 11 units left and 8 units down, and a reflection across the x -axis.
- c A translation 11 units left and 8 units down, and a rotation by 90° counterclockwise about the origin.
- d A rotation by 90° counterclockwise about the origin, and a translation 11 units left and 8 units down.



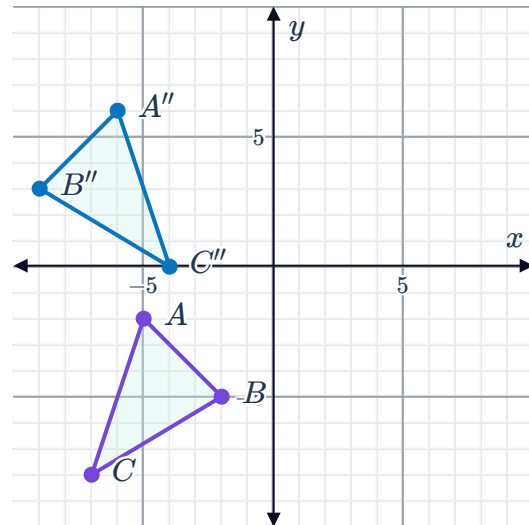
30 Determine whether the following sequences of transformations map quadrilateral $ABCD$ to quadrilateral $A''B''C''D''$.

- a A translation 12 units left and 11 units up, and a rotation by 270° clockwise about the origin.
- b A rotation by 270° clockwise about the origin, and a translation 12 units left and 11 units up.
- c A translation 12 units left and 11 units up, and a reflection across the y -axis.
- d A reflection across the y -axis, and a translation 12 units left and 11 units up.



31 Determine whether the following sequence of transformations map $\triangle ABC$ to $\triangle A''B''C''$.

- a A rotation by 180° counterclockwise about the origin, and a reflection across the x -axis.
- b A reflection across the x -axis, and a translation 11 units right and 8 units up.
- c A translation 11 units right and 8 units up, and a reflection across the y -axis.
- d A reflection across the y -axis, and a rotation by 180° counterclockwise about the origin.



32 Consider the following diagram.

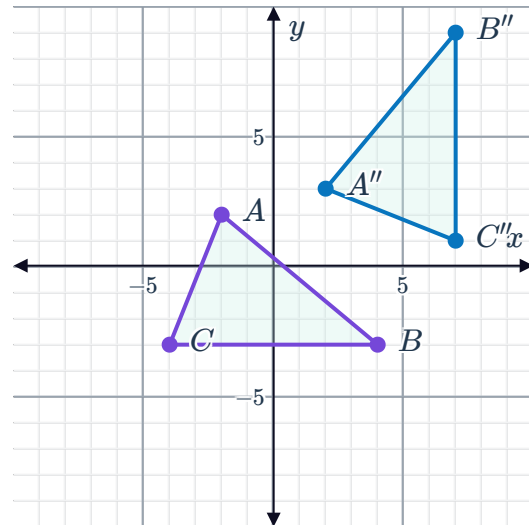
Complete the tables below to identify the change in coordinates going through the sequence of transformations from $\triangle ABC$ to $\triangle A''B''C''$.

- a A translation 5 units to the right and 4 units down.

$A(-2, 2)$	$\rightarrow$	$A'(\quad, \quad)$
$B(4, -3)$	$\rightarrow$	$B'(\quad, \quad)$
$C(-4, -3)$	$\rightarrow$	$C'(\quad, \quad)$

- b A rotation 270° clockwise about the origin.

$A'(\quad, \quad)$	$\rightarrow$	$A''(\quad, \quad)$
$B'(\quad, \quad)$	$\rightarrow$	$B''(\quad, \quad)$
$C'(\quad, \quad)$	$\rightarrow$	$C''(\quad, \quad)$



33 Consider the following diagram.



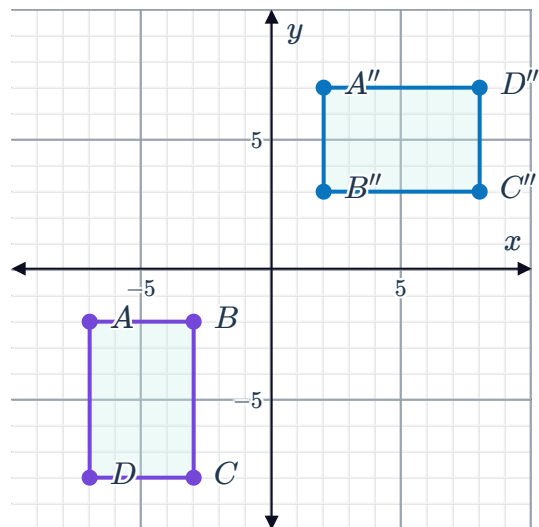
Complete the tables below to identify the change in coordinates going through the sequence of transformations from quadrilateral $ABCD$ to quadrilateral $A''B''C''D''$.

- a** A rotation by 90° clockwise about the origin.

$A(-7, -2)$	$\rightarrow$	$A'(\quad, \quad)$
$B(-3, -2)$	$\rightarrow$	$B'(\quad, \quad)$
$C(-3, -8)$	$\rightarrow$	$C'(\quad, \quad)$
$D(-7, -8)$	$\rightarrow$	$D'(\quad, \quad)$

- b** A reflection across the y -axis.

$A'(\quad, \quad)$	$\rightarrow$	$A''(\quad, \quad)$
$B'(\quad, \quad)$	$\rightarrow$	$B''(\quad, \quad)$
$C'(\quad, \quad)$	$\rightarrow$	$C''(\quad, \quad)$
$D'(\quad, \quad)$	$\rightarrow$	$D''(\quad, \quad)$

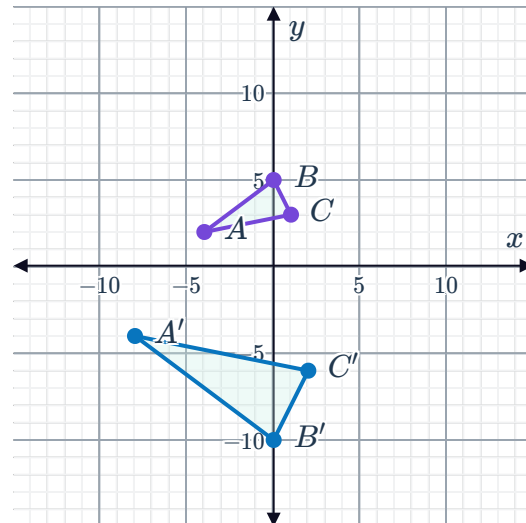


34 Consider the quadrilateral with vertices at $(5, 4)$, $(0, 4)$, $(0, -1)$, and $(5, -1)$, and the quadrilateral with vertices at $(4, 5)$, $(4, 0)$, $(-1, 0)$, and $(-1, 5)$. Determine whether the following sequences of transformations maps the first quadrilateral to the second quadrilateral.

- a** Rotated 270° clockwise about the origin and reflected across the x -axis.
- b** Reflected across the x -axis and rotated 270° clockwise about the origin.
- c** Reflected across the x -axis and translated 2 units right and 3 units up.
- d** Translated 2 units right and 3 units up and reflected across the x -axis.

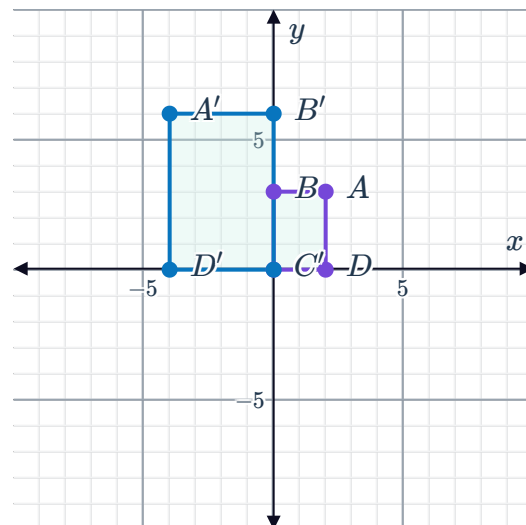
35 Determine whether the following sequence of transformations maps $\triangle ABC$ to $\triangle A'B'C'$.

- a Translated 3 units right and 4 units down, and dilated by a factor of 2 with the origin as the center of dilation.
- b Rotated 90° clockwise around the origin and reflected about the x -axis.
- c Reflected about the x -axis and dilated by a factor of 2 with the origin as the center of dilation.
- d Rotated 90° clockwise around the origin and dilated by a factor of 2 with the origin as the center of dilation.



36 Determine whether the following sequences of transformations maps quadrilateral $ABCD$ to quadrilateral $A'B'C'D'$.

- a Reflected about the y -axis and rotated 180° clockwise around the origin.
- b Translated 3 units left and 2 units down, and dilated by a factor of 2 with the origin as the center of dilation.
- c Rotated 180° clockwise around the origin and dilated by a factor of 2 with the origin as the center of dilation.
- d Dilated by a factor of 2 with the origin as the center of dilation and reflected about the y -axis.



- 37** Consider the triangle with vertices at $A(-4, 5)$, $B(-1, 5)$, and $C(1, 3)$, and the triangle with vertices at $A'(8, 4)$, $B'(2, 10)$, and $C'(-2, 6)$.



Determine whether the following sequences of transformations maps the first triangle to the second triangle.

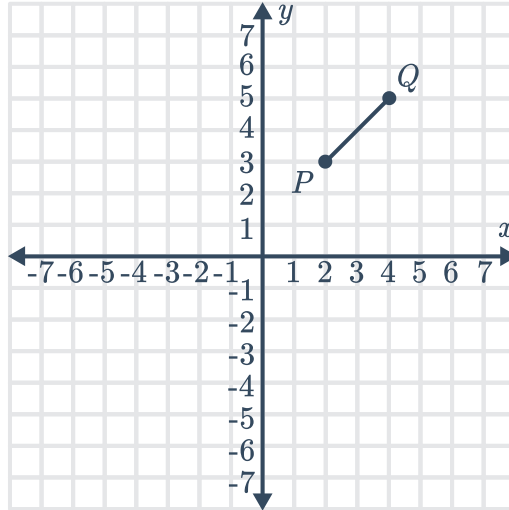
- a** Dilated by a factor of 2 with the origin as the center of dilation and rotated 180° counterclockwise around the origin.
 - b** Rotated 180° counterclockwise around the origin and translated 5 units right and 4 units down.
 - c** Dilated by a factor of 2 with the origin as the center of dilation, and translated 5 units right and 4 units down.
 - d** Reflected about the y -axis and dilated by a factor of 2 with the origin as the center of dilation.
- 38** Consider the quadrilateral with vertices at $(3, 2)$, $(-1, 2)$, $(-1, -3)$, and $(3, -3)$, and the quadrilateral with vertices at $(-6, 9)$, $(-6, -3)$, $(9, -3)$, and $(9, 9)$. Determine whether the following sequences of transformations maps the first quadrilateral to the second quadrilateral.
- a** Dilated by a factor of 3 with the origin as the center of dilation and rotated 90° counterclockwise around the origin.
 - b** Reflected about the y -axis and translated 2 units left and 3 units down.
 - c** Reflected about the y -axis and dilated by a factor of 3 with the origin as the center of dilation.
 - d** Dilated by a factor of 3 with the origin as the center of dilation, and translated 2 units left and 2 units down.
- 39** The quadrilateral with vertices at $(9, 9)$, $(0, 9)$, $(0, 0)$, and $(9, 0)$ is rotated 90° clockwise around the origin and dilated by a factor of 2 with the origin as the center of dilation. Determine the new coordinates of the vertices of the quadrilateral.



Additional questions

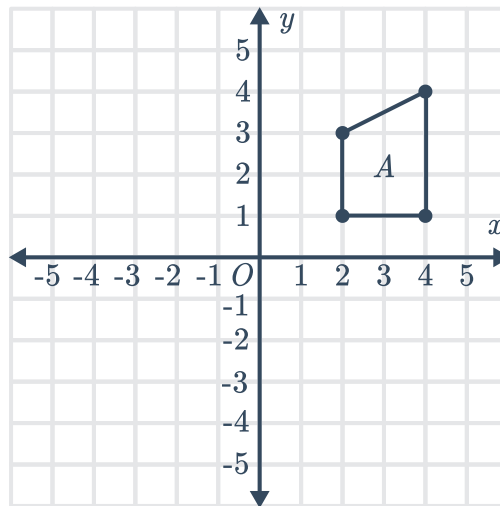
40 Perform the following sequence of transformations on $\overline{PQ}$:

- Rotate 90° counterclockwise about the origin
- Translate 5 units down

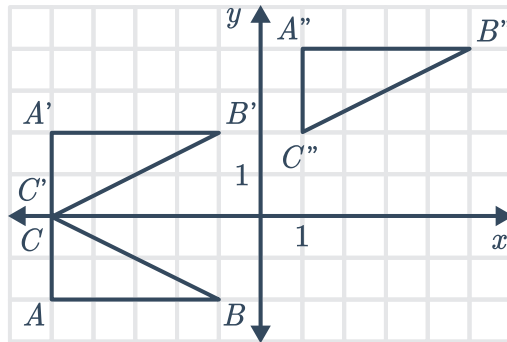


41 Perform the following sequence of transformations on Figure A:

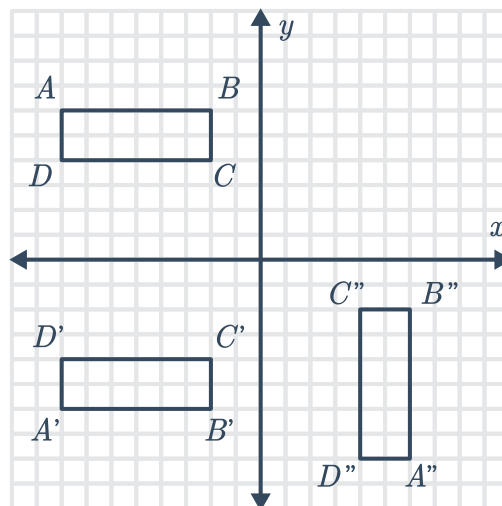
- Reflection across the y -axis
- Rotation of 180° clockwise about the origin



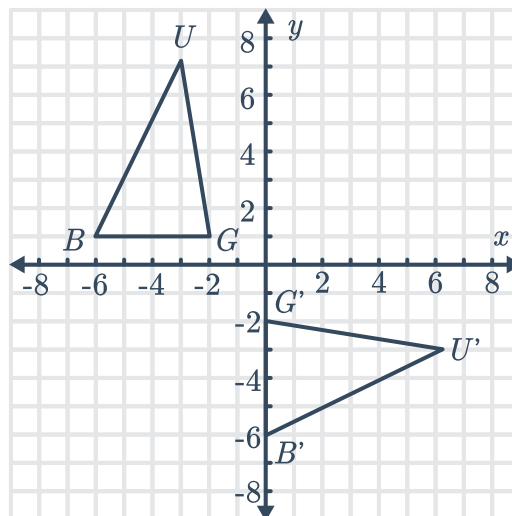
42 Describe a sequence of transformations that maps $\triangle ABC$ to $\triangle A''B''C''$:



43 Describe a sequence of transformations that maps $ABCD$ to $A''B''C''D''$:



44 Describe a sequence of transformations that maps $\triangle BUG$ to $\triangle B''U''G''$:



- 45 The endpoints of $\overline{JK}$ are $(9, 1)$, $(10, 3)$.



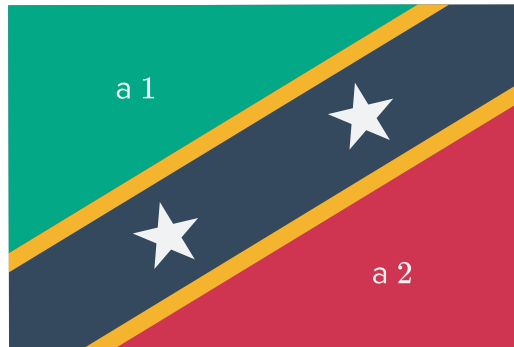
Graph the image of $\overline{JK}$ after rotation by 180° about the origin then reflection across the x -axis.

- 46 The vertices of quadrilateral $JKLM$ are $(-4, -2)$, $(0, 8)$, $(5, 0)$, and $(5, -1)$.

Graph the image of $JKLM$ after the following sequence of transformations:

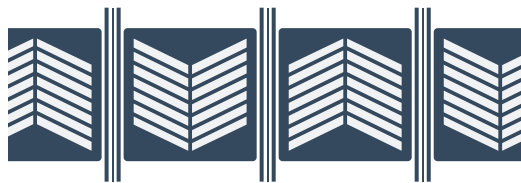
- Rotate 180° counterclockwise
- Reflect across $y = 1$

- 47 Here is the flag for Trinidad and Tobago:



- a Describe how to map section 1 of the flag onto section 2 using a single transformation.
- b Describe how to map section 1 of the flag onto section 2 using two consecutive transformations.

- 48 Here is pattern from a piece of Kente fabric, woven in Ghana:



Describe a sequence of transformations that could be used to recreate the pattern.